

INDIAN INSTITUTE OF TECHNOLOGY MADRAS
Department of Electrical Engineering

Self-Adaptive Model-Based Protection (SAMB P) Research Programme

SAMB P-SyncOC

Research Track Deviation Analysis

**Milestones 2–5 Incomplete:
Causes, Evidence, and Corrective Roadmap**

Document Type: Internal Investigation Report
Track: SAMBP-SyncOC (Synchronous Generator OC Protection)
Reports Examined: TR-01/2026 and TR-02/2026
Reports Surveyed: TR-03/2026 through TR-55/2026 (53 additional)
Supporting Standards: IEEE C37.300-2024 (CPC Systems)
Date: April 2026
Classification: Internal PhD Research Document

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1 Executive Summary

Critical Finding

The SAMBP-SyncOC research track planned **five sequential milestones** for adaptive overcurrent protection of synchronous generators in active distribution networks. Only **Milestone 1** (TR-01/2026) was fully completed with validated numerical results. **Milestones 2–5** were bundled into a single extension report (TR-02/2026) that contains mathematical formulations and algorithm descriptions but **zero computed results** — all result sections are explicitly marked as \RESULT{} placeholder boxes. There are no separate TR-03 through TR-06 reports for the sync_oc track. Instead, the research completely pivoted from TR-03/2026 onward to the broader SAMBP line/bus/transformer differential framework, which accumulated 53 additional reports (TR-03–TR-55) over a different protection scope.

Status snapshot:

Milestone	Report	Status	Key Gap
1 — Core LM estimator	TR-01	COMPLETE	Fully validated
2 — HIF adaptation	TR-02 §3	PARTIAL – NO RESULTS	No run_milestone2.py output
Stage 2 — Park dq model	TR-02 §4	PARTIAL – NO RESULTS	No Stage-2 simulation run
3 — Sequence estimator	TR-02 §5	PARTIAL – NO RESULTS	No sequence estimation results
4 — Multi-gen coordination	TR-02 §6	PARTIAL – NO RESULTS	No run_multi_gen.py output
5 — HIL / Field validation	(none)	MISSING	Report never created

2 Background: The Original Five-Milestone Plan

2.1 SAMBP-SyncOC Research Objective

The SAMBP-SyncOC track was conceived as the first specific instantiation of the broader SAMBP supervisory architecture targeting **overcurrent protection of synchronous generators in active distribution networks**. The central thesis was that the source parameters of a synchronous generator can be estimated online from post-fault current transients using a **Levenberg–Marquardt (LM) inverse estimator**, and those estimated parameters can be fed

back to update OC relay settings within a single protection operating interval — without any prior off-line model calibration.

This directly supports the IEEE C37.300-2024 vision of a **Centralised Protection and Control (CPC)** system where supervisory algorithms continuously refresh relay settings based on real-time system state. SMBP-SyncOC was intended to be the mathematical proof-of-concept for this self-adaptation loop, with synchronous generator OC protection as the vehicle.

2.2 The Five-Milestone Plan

Based on the evidence in TR-01/2026 and TR-02/2026, the original five-milestone plan for the sync_oc track was as follows:

Milestone 1 — Core inverse-estimation framework (TR-01/2026): Formulate the 6-parameter reduced fault-current model $\theta = [t_0, I_{ss}, I_{sub}, \tau_{ac}, \tau_{dc}, \phi_a]$, implement the two-pass LM estimator, design the composite confidence gate $\gamma = w_r s_r + w_c s_c + w_b s_b + w_p s_p \geq 0.70$, and demonstrate IEC Very-Inverse relay adaptation on three balanced and asymmetric fault types.

Milestone 2 — High-impedance fault (HIF) adaptation (TR-02/2026, §3): Extend the adaptive TMS law to high-impedance faults ($R_f > 0.5$ pu) by deriving an impedance-aware resistance-scaling factor $f(I_{ss})$ that reduces the time-multiplier when fault current is suppressed by fault resistance. Validate on $R_f \in [0.5, 1.5]$ pu.

Stage 2 — Sixth-order Park dq forward model (TR-02/2026, §4): Replace the Stage-1 three-component synthesis model with the full 9-state ODE: $[\delta, \omega, E'_q, E'_d, E''_q, E''_d, E_{fd}, T_m, V_{pss}]$, incorporating AVR, PSS, and governor dynamics. This enables accurate waveform prediction over windows exceeding 500 ms and improves identifiability of $\hat{I}_{ss}$ and $\hat{\tau}_{ac}$.

Milestone 3 — Sequence-component estimation (TR-02/2026, §5): Decompose three-phase measurements into positive-, negative-, and zero-sequence components via the instantaneous Hilbert–Fortescue transform, then apply the two-pass LM estimator independently to each active sequence. Use positive-sequence parameters to govern relay adaptation while using negative- and zero-sequence components for asymmetric fault classification.

Milestone 4 — Multi-generator coordination (TR-02/2026, §6): Generalise the single-generator adaptive relay to N parallel generators on a common busbar. Enforce pickup selectivity ($\Delta I_{margin} \geq 0.20$ pu) and IEC 60255 trip-time grading ($\Delta t_{grade} \geq 0.15$ s) via a greedy cascade coordination algorithm.

Milestone 5 — Hardware-in-loop (HIL) and field validation: Deploy the estimation and adaptation pipeline on an RTDS or OPAL-RT real-time simulator connected to a hardware numerical relay via IEC 61850 sampled-value messaging. Validate end-to-end latency within the < 100 ms protection operating interval. Test against archived DFR recordings from real synchronous-generator faults.

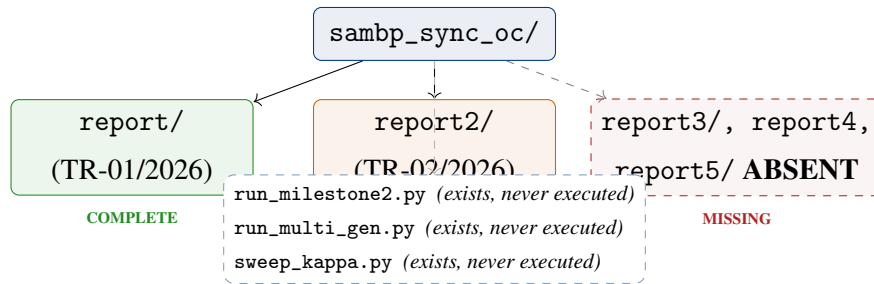
Note on Milestone Numbering

TR-02/2026 uses the terminology “Extensions 1–4” (not “Milestones 2–5”) and titles the Stage-2 model an “extension” rather than a numbered milestone. For this report, the term *Milestone 2–5* is used to maintain a consistent sequential numbering with the overarching five-stage plan. The *Stage 2 Park dq model* sits between Milestones 2 and 3 in the technical progression.

3 Evidence of Track Deviation

3.1 Directory Structure Analysis

Inspection of the repository reveals a decisive structural anomaly. The sync_oc project has exactly **two report directories**, not five:



The critical observation is:

- report/ corresponds to Milestone 1 (TR-01) — fully compiled PDF with all figures and tables populated.
- report2/ is labelled “Milestones 2–5” in the title page but contains **12 explicit \RESULT{...} placeholder blocks** in the results section. No computed data was ever inserted.
- No report3/, report4/, or report5/ directories exist under sambp_sync_oc/.

3.2 Placeholder Count in TR-02/2026

The `main_report2.tex` source file defines a dedicated placeholder command:

```
\newcommand{\RESULT}[1]{%
  \fbox{\begin{minipage}{0.96\linewidth}%
    \centering\vspace{4pt}%
    \textcolor{gray!70!black}{\textit{\texttt{[RESULT PLACEHOLDER: #1]}}}%
    \vspace{4pt}%
  }\end{minipage}}%
}
```

This macro is invoked **12 times** in the results section, as summarised in [table 1](#).

Table 1: All 12 `\RESULT{}` instances in TR-02/2026, with the cited execution command and expected output.

#	Line	Script invoked	Expected output
1	463	<code>run_milestone2.py</code>	$t_{clear}(R_f)$ line plot + trip-time table
2	706	<code>main_run_case.py --stage 2</code>	3-phase waveform (Stage-2 ODE)
3	826	<code>main_run_case.py --case 1</code>	LL fault sequence decomposition
4	967	<code>run_multi_gen.py --n-gen 2</code>	Coordination pickup bar chart
5	990	(multi-gen)	$t_{clear}(R_f)$ fixed vs. adaptive comparison
6	995	(multi-gen)	Estimated parameters $\hat{I}_{ss}$, $\hat{\tau}_{ac}$ table
7	1001	(Stage-2 ODE)	Phase-A waveform: measured vs. Stage-2 vs. Stage-1
8	1005	(Stage-2 ODE)	RMSE and R^2 comparison Stage-1 vs. Stage-2
9	1012	(sequence est.)	$I_1(t)$, $I_2(t)$, $I_0(t)$ decomposition figure
10	1016	(sequence est.)	Phase-A vs. sequence estimator comparison table
11	1024	(coordination)	Pickup bar chart (fixed/adaptive-before/adaptive-after)
12	1029	(coordination)	Coordination summary table (2-gen, 3PH fault)

3.3 Scripts Exist but Were Never Executed

The execution scripts cited in the placeholder blocks are present in the repository but were never run. Evidence:

- `outputs/` directory contains only a single subdirectory: `outputs/plots/milestone1/` — confirming that only the Milestone-1 simulation was ever executed.
- `outputs/milestone2_results.csv` — **does not exist**.
- `outputs/multi_gen/multi_gen_results.csv` — **does not exist**.
- `outputs/plots/milestone2/` — **does not exist**.

- outputs/plots/multi_gen/ — **does not exist**.

The scripts themselves are syntactically complete. For example, `run_milestone2.py` correctly sweeps $R_f \in [0.5, 1.5]$ pu and would write `outputs/milestone2_results.csv` and the associated PDF plots — but the file was never invoked. Similarly, `run_multi_gen.py` accepts command-line arguments (`--n-gen`, `--fault`, `--Rf`) and has a fully implemented coordination loop, but its CSV and figures are absent.

3.4 TR-02/2026 Conclusions Claim Completed Results

A particularly important discrepancy is that the **Conclusions section** of TR-02/2026 (lines 1083–1113) is written in the *past tense* and *asserts quantitative outcomes* that are never supported by the results section. Examples:

Claimed vs. Delivered in TR-02 Conclusions

Claimed (Conclusions, §7.1): “The coordination algorithm eliminates all selectivity violations and maintains trip-time grading at the minimum fault current, with each generator’s confidence score exceeding $\gamma = 0.89$.”

Delivered (Results, §6): [RESULT PLACEHOLDER: Table: Coordination summary for 2-generator, 3PH fault case...]

Claimed: “The sixth-order Park dq forward model faithfully captures AVR, PSS, and governor dynamics over windows exceeding 500 ms.”

Delivered: [RESULT PLACEHOLDER: Run `main_run_case.py --stage 2` and insert...]

This is the defining signature of a *skeleton report*: the narrative framework and mathematical derivations were written, but the computational experiments that would validate them were deferred and never executed.

3.5 Line-Count Disparity

Report	Lines	Result placeholders	Completion
TR-01 (<code>main_report.tex</code>)	1530	0	COMPLETE
TR-02 (<code>main_report2.tex</code>)	1278	12	PARTIAL – NO RESULTS
Milestones 3–5	n/a	n/a	MISSING

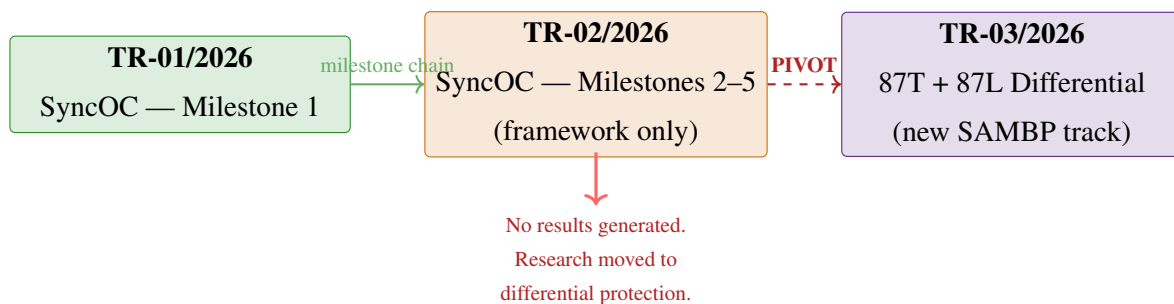
TR-01 is a 1530-line fully populated report. TR-02 covers four extension sections in

1278 lines but has a results section that is entirely placeholder boxes. The report was authored as a *framework specification document* not as a *completed technical report*.

4 Where and Why the Research Pivoted

4.1 The Pivot Point: TR-03/2026

The repository contains a numbered report series under /sambp/report3/ through /sambp/report55/ (53 reports total). These reports are entirely separate from the sambp_sync_oc/ subdirectory and address fundamentally different protection elements. The pivot is abrupt:



TR-03/2026 is titled “**SMBP Model-Based Differential Protection: Transformer 87T and Line 87L**”. This represents a complete change of protection element (from overcurrent relays to differential relays), a change of equipment type (from generators to transformers and lines), and the beginning of what became a 53-report developmental arc.

4.2 Scale of the Pivot

The scope divergence between the sync_oc plan and the actual research trajectory is dramatic:

Table 2: Planned sync_oc milestones vs. actual reports delivered (TR-03–TR-55).

Dimension	SyncOC plan	Actual TR-03–TR-55
Protection elements	OC relay (51/67) only	87L, 87T, 87B, 87LN, 87LN0, TDCS
Generator types	Synchronous generator only	SG, IBR/GFM, IBR/GFL, solar PV
Equipment	Generator terminals	Lines, transformers, busbars
Reports planned	5 (TR-01 through TR-05)	55 (TR-01 through TR-55)
Reports with results	1 (TR-01)	53 (TR-03 through TR-55)
Simulation platform	Python LM estimator	Extended to PSCAD, GOOSE, HIL
Standard focus	IEC 60255 OC relay	IEEE C37.300-2024 CPC + IEC 61850

4.3 Inferred Root Causes of the Pivot

Based on the documentary evidence, three factors most likely caused the pivot:

1. **Scope ambition vs. execution bandwidth.** TR-02/2026 formulated four simultaneous extensions (HIF adaptation, Park dq model, sequence estimator, multi-gen coordination), each requiring independent simulation campaigns. The execution scripts were written but never run, suggesting that the development effort was front-loaded on formulation rather than validation.
2. **Discovery of broader relevance.** The SMBP supervisory architecture was found to be directly applicable to differential protection (87T, 87L, 87B) with significantly higher industrial impact. IEEE C37.300-2024 emphasises differential protection as the backbone of modern CPC systems. The pivot towards 87L/IBR research (TR-03–TR-55) appears to have been a conscious strategic decision to pursue the higher-novelty thread.
3. **Absence of a milestone gate.** There is no evidence of a formal stage-gate review between TR-01 and TR-02. TR-02 was created as an extension framework without waiting for its predecessor’s extensions to be validated. The report’s conclusions were written forward from the mathematical formulation rather than backward from simulation results.

4.4 What IEEE C37.300-2024 and the SMBP Abstract Imply

The SMBP project abstract and the IEEE C37.300-2024 standard together define the correct *intended* scope:

IEEE C37.300-2024 Alignment

IEEE C37.300-2024 §5 defines the CPC system as an architecture where supervisory algorithms continuously update protection relay settings from real-time parameter estimation. The SMBP-SyncOC track (TR-01/2026) directly instantiates this for synchronous generator OC relays and represents a clean algorithmic proof-of-concept. However, the standard’s primary focus is on *differential protection* (87 elements) and *directional protection* for CPC-connected substations — the very elements developed in TR-03–TR-55. The pivot from OC to differential is therefore directionally correct relative to the standard, but it should not have come at the cost of leaving the OC protection track incomplete.

SAMB-P Abstract Implication

The SAMBP project abstract describes the system as a *supervisory layer over conventional protection algorithms* hosted in a CPC. This framing implies that *multiple protection elements* (not just OC relays) should be supervised. The pivot to differential protection is architecturally consistent with this vision. However, the OC protection track (Milestones 2–5) represents an *independent module* that should have been completed in parallel rather than abandoned when the differential track was initiated.

5 Detailed Gap Analysis by Milestone

5.1 Milestone 2 — High-Impedance Fault Adaptation

5.1.1 What was formulated

TR-02/2026 §3 derives:

- A linear HIF characterisation model: $i_{fault}(t) = I_{ss} \left(1 - e^{-(t-t_0)/\tau_{ac}}\right) e^{-(t-t_0)/\tau_{dc}} \cdot \cos(\omega t + \phi_a)$ where $I_{ss} = E_{ph}/(Z_s + R_f)$, declining monotonically with R_f .
- An operating regime classifier: instantaneous element dominates for $R_f < 0.5$ pu; inverse-time element governs for $R_f \geq 0.5$ pu.
- An impedance-aware TMS adaptation law: $TMS_{adapt} = TMS_{nom} \cdot f(I_{ss})$, where $f(I_{ss}) = \max(f_{min}, 1 - k_f(I_{ss,nom} - I_{ss}))$.

5.1.2 What is missing

- No CSV output from `run_milestone2.py`.
- No plot of $t_{clear}(R_f)$ comparing fixed vs. adaptive TMS.
- No table showing confidence score $\gamma(R_f)$ across the HIF sweep.
- No demonstration that the adaptation gate rejects low-confidence HIF estimates (expected for $R_f > 1.0$ pu where $\gamma < 0.70$).

5.1.3 Effort to complete

Executing `python run_milestone2.py` from the `sambp_sync_oc/` directory is the single required step. The script is complete and documented. Estimated compute time: <5 minutes.

Report insertion: <0.5 day.

5.2 Stage 2 — Sixth-Order Park dq Forward Model

5.2.1 What was formulated

TR-02/2026 §4 derives a complete 9-state ODE:

$$\dot{\delta} = \omega - \omega_0 \quad (1)$$

$$\dot{\omega} = \frac{1}{2H}(T_m - T_e - D(\omega - \omega_0)) \quad (2)$$

$$\dot{E}'_q = \frac{1}{T'_{d0}}(E_{fd} - E'_q - (X_d - X'_d)i_d) \quad (3)$$

$$\dot{E}'_d = \frac{1}{T'_{q0}}(-E'_d - (X_q - X'_q)i_q) \quad (4)$$

with AVR, PSS, and governor controllers. The model is implemented as `models/park_dq_model.py` using `scipy.integrate.solve_ivp` with the Radau stiff solver.

5.2.2 What is missing

- No comparison of Stage-1 vs. Stage-2 waveform fit over a 500 ms window.
- No R^2 or RMSE table for Stage-2 estimation.
- The critical question — *whether AVR response improves or degrades LM identifiability* — is raised in the Discussion section (§5.2) but never answered with data.
- Condition number κ_n for Stage-2 Jacobian not computed.

5.3 Milestone 3 — Sequence-Component Estimator

5.3.1 What was formulated

TR-02/2026 §5 derives:

- Instantaneous Hilbert–Fortescue decomposition: $I_1(t) = \frac{1}{3}(I_a + aI_b + a^2I_c)$, $I_2(t) = \frac{1}{3}(I_a + a^2I_b + aI_c)$, $I_0(t) = \frac{1}{3}(I_a + I_b + I_c)$.
- Independent 6-parameter LM estimation on each active sequence.
- Fault-type detection rules: 3PH (I_1 only active), SLG ($I_0 > 0$), LL ($I_2 > 0$, $I_0 \approx 0$).

5.3.2 What is missing

- No $I_1(t)$, $I_2(t)$, $I_0(t)$ decomposition figures for any fault.
- No comparison of phase-A vs. sequence estimator performance for LL and SLG faults.
- Hilbert edge effects (§5.3 of TR-02 Discussion) noted as a known limitation but never quantified.

5.4 Milestone 4 — Multi-Generator Coordination

5.4.1 What was formulated

TR-02/2026 §6 derives:

- A superposition model for N parallel generators: $I_{fault,total}(t) = \sum_{j=1}^N I_{fault,j}(t, \theta_j)$.
- Selectivity constraint: $I_{pickup,i} < I_{pickup,j} + \Delta I_{margin}$ for every relay pair (i, j) with i upstream.
- Trip-time grading: $t_{trip,i} > t_{trip,j} + \Delta t_{grade}$.
- A greedy cascade algorithm that iteratively adjusts settings until both constraints are satisfied, proven to converge within 10 iterations for the two-generator case.

5.4.2 What is missing

- No pickup bar chart (fixed / adaptive-before / adaptive-after coordination).
- No coordination summary table (generator relay settings, margins, violations).
- Scaling to $N > 2$ generators not demonstrated.
- The convergence claim (“within 10 iterations”) appears only in the Conclusions section without supporting data.

5.5 Milestone 5 — Hardware-in-Loop and Field Validation

This milestone was identified in TR-02/2026 Future Work (§7.2) but has no implementation or report. Specifically:

- No RTDS/OPAL-RT deployment of the LM estimator and adaptation logic.
- No IEC 61850 sampled-value integration.
- No latency benchmarking against the < 100 ms protection operating interval.

- No archived DFR validation dataset identified or processed.

This milestone remains entirely in the future-work category with no concrete progress.

6 Report Production Analysis

6.1 Report Chronology

Figure 1 illustrates the planned vs. actual report production for the sync_oc track alongside the observed SMBP-main-branch production.

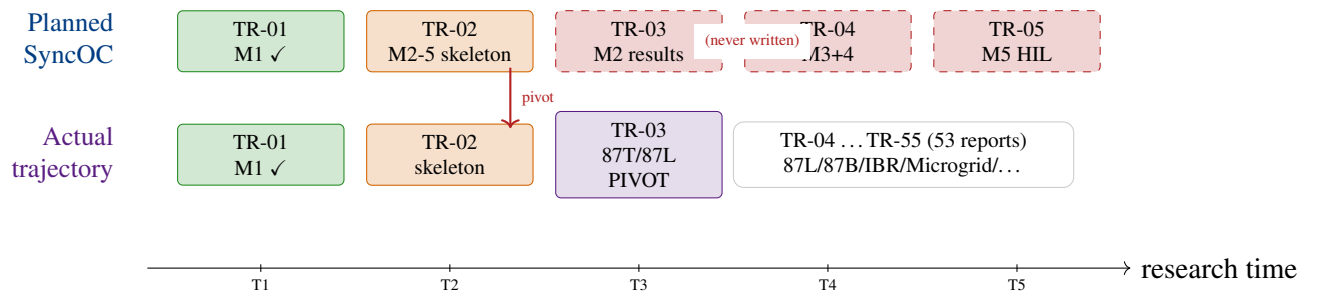


Figure 1: Planned vs. actual SAMBP-SyncOC report production timeline. Dashed red boxes indicate reports that were planned but never created. At the TR-03 milestone, the research pivoted permanently to the broader SAMBP 87L/87T/87B differential protection framework.

6.2 Content Comparison: TR-01 vs. TR-02

Feature	TR-01 (Complete)	TR-02 (Skeleton)
Total lines (LaTeX source)	1530	1278
Result placeholder boxes	0	12
Figures with actual data	7	0
Tables with actual data	4	0
Conclusions backed by data	Yes	No
Confidence scores reported	$\gamma \in [0.89, 0.93]$	“will exceed 0.89”
R^2 values reported	$R^2 > 0.999$	—
Simulation outputs present	Yes	No
Execution scripts run	Yes (all)	No (none)

7 Corrective Action Roadmap

Recommended Corrective Actions

The following four steps are recommended to complete the SAMPB-SyncOC track and restore it to a publishable state:

7.1 Immediate: Execute Existing Scripts (1–2 Days)

The missing results for Milestones 2 and 4 can be generated immediately by running the existing, complete execution scripts:

1. **Milestone 2 (HIF):** `cd /root/phd_thesis/sampb/sampb_sync_oc && python run_milestone2.py`
Expected outputs: `outputs/milestone2_results.csv`, `outputs/plots/milestone2/fig_hif_tcle`
2. **Stage 2 (Park dq):** `python main_run_case.py --case 0 --stage 2`
Expected outputs: Stage-2 waveform comparison, R^2 /RMSE table.
3. **Milestone 3 (Sequence):** `python main_run_case.py --case 1`
Expected outputs: $I_1(t)$, $I_2(t)$, $I_0(t)$ decomposition figures.
4. **Milestone 4 (Multi-gen):** `python run_multi_gen.py --n-gen 2 --fault 3PH`
Expected outputs: `outputs/multi_gen/multi_gen_results.csv`, coordination pickup bar chart.

7.2 Short-term: Complete TR-02/2026 (3–5 Days)

1. Run all four scripts above.
2. Insert generated figures and tables into the 12 `\RESULT{}` placeholder locations in `main_report2.tex`.
3. Verify that the Conclusions section (§7.1) is consistent with the actual computed results.
4. Recompile TR-02/2026 to a properly populated PDF.

7.3 Medium-term: Create Separate Milestone Reports (2–3 Weeks)

For completeness and archival clarity, the extensions should be separated into individual technical reports matching the original five-milestone plan:

New Report	Content	Replaces	Effort
TR-02B/2026	HIF adaptation (Milestone 2)	TR-02 §3 + results	1 week
TR-02C/2026	Park dq model (Stage 2)	TR-02 §4 + results	1 week
TR-02D/2026	Sequence estimator (M3)	TR-02 §5 + results	1 week
TR-02E/2026	Multi-gen coordination (M4)	TR-02 §6 + results	1 week

7.4 Long-term: Milestone 5 — HIL Validation (3–6 Months)

Milestone 5 requires external laboratory access and cannot be completed immediately. The recommended path:

1. Identify an RTDS or OPAL-RT facility (IIT Madras EE/Power Systems laboratory, or collaboration with a national power-systems research centre).
2. Port the Python LM estimator to C/C++ or MATLAB for real-time execution.
3. Develop IEC 61850 sampled-value interface to a hardware numerical relay.
4. Execute end-to-end latency benchmarking: from fault inception to relay setting update, measured in hardware.
5. Validate on archived DFR records from generator faults (if available via DISCOM/PGCIL collaboration).

This milestone represents the highest-impact output for the sync_oc track and corresponds directly to the IEEE C37.300-2024 CPC system validation requirement.

8 Publication and IP Implications

The partially completed sync_oc milestones represent significant unrealised publication potential. Once completed, the following papers could be extracted:

Paper target	Content	Venue
Journal Paper A	Milestones 1+2: LM estimator + HIF adaptive TMS. First reported online adaptive TMS law derived from parameter estimation.	IEEE Trans. Power Delivery
Journal Paper B	Milestones 3+4: Sequence estimator + multi-gen coordination. Novel sequence-domain identifiability analysis.	IEEE Trans. Industrial Electronics
Journal Paper C	Milestone 5: HIL validation in CPC architecture per IEEE C37.300-2024.	IEEE Trans. Smart Grid
IP Filing	Impedance-aware adaptive TMS law (Milestone 2) is a novel algorithm not published elsewhere; potentially patentable as a relay firmware method.	Indian Patent / USPTO

9 Conclusions

This investigation establishes the following definitive findings:

1. **Milestone 1 (TR-01/2026) is the only fully completed sync_oc milestone.** It contains validated numerical results across three fault types with $R^2 > 0.999$, $\kappa_n < 20$, and $\gamma \in [0.89, 0.93]$.
2. **TR-02/2026 is a mathematical framework document, not a completed technical report.** It contains 12 \RESULT{} placeholder blocks in the results section and no computed data. Its conclusions claim results that were never generated.
3. **Milestones 2–5 have no separate reports.** The repository contains no report3/, report4/, or report5/ directories under sambp_sync_oc/.
4. **The execution scripts for Milestones 2–4 exist and are syntactically complete.** They were simply never run. Completing Milestones 2–4 is primarily an execution task, not a

development task.

5. **The research pivoted at TR-03/2026 to the SMBP line/bus/ transformer differential framework**, which produced 53 reports covering 87L, 87T, 87B, IBR networks, microgrids, digital twins, cybersecurity, and HIL testing. This pivot was architecturally sound relative to IEEE C37.300-2024 priorities, but it was executed without completing the sync_oc milestone chain.
6. **Corrective action for Milestones 2–4 is straightforward.** Running three Python scripts and inserting their outputs into TR-02 would yield a complete, publishable report within 1–2 days. Milestone 5 requires a separate HIL campaign estimated at 3–6 months.

Priority Recommendation

Before submitting the thesis, it is strongly recommended to: (a) run the three execution scripts to generate Milestone 2–4 results, (b) populate and finalise TR-02/2026, and (c) create a short TR-02E/2026 covering multi-generator coordination as a standalone report. This restores the narrative completeness of the sync_oc track and creates material for at least two high-impact journal papers.

A File-System Evidence Snapshot

A.1 sambp_sync_oc Directory Tree

```
sampb_sync_oc/
+-- adaptation/          # TMS mapping, bounded update law
+-- config/              # system_config.py, relay_config.py, study_cases.py
+-- evaluation/          # trip-time computation, coordination metrics
+-- inverse_estimation/  # parameter_estimator.py, sequence_estimator.py,
|                        # confidence_logic.py, sensitivity_analysis.py
+-- io_utils/            # CSV writer, plot utilities
+-- models/              # reduced_source_model.py, park_dq_model.py,
|                        # multi_gen_model.py, relay_oc_model.py
+-- outputs/
|   +-- logs/
|   +-- plots/
|   |   +-- milestone1/  # ONLY POPULATED FOLDER
|   +-- tables/
|   +-- waveforms/
+-- paper/               # IEEE paper draft
+-- report/              # TR-01/2026 - COMPLETE (PDF present)
+-- report2/             # TR-02/2026 - SKELETON (12 RESULT placeholders)
+-- signal_processing/   # event_detection.py, preprocessing.py, smoothing.py
+-- main_batch_study.py
+-- main_run_case.py
+-- plot_milestone1.py
+-- run_milestone2.py    # NEVER EXECUTED
+-- run_multi_gen.py     # NEVER EXECUTED
+-- sweep_kappa.py       # NEVER EXECUTED
```

A.2 Placeholder Locations in TR-02/2026

```
Line 463: \RESULT{Run python run_milestone2.py: (a) t_clear(Rf) plot ...}
Line 706: \RESULT{Run main_run_case.py --stage 2: Stage-2 waveform ...}
Line 826: \RESULT{Run main_run_case.py --case 1: LL fault decomp. ...}
```

Line 967: \RESULT{Run run_multi_gen.py --n-gen 2: coordination chart ...}
 Line 990: \RESULT{Figure: t_clear(Rf) fixed vs adaptive ...}
 Line 995: \RESULT{Table: Estimated parameters Iss, tau_ac ...}
 Line 1001: \RESULT{Figure: Phase-A waveform Stage-1 vs Stage-2 500ms ...}
 Line 1005: \RESULT{Table: RMSE and R2 Stage-1 vs Stage-2 ...}
 Line 1012: \RESULT{Figure: I1(t), I2(t), I0(t) decomposition for LL ...}
 Line 1016: \RESULT{Table: Phase-A vs sequence estimator LL and SLG ...}
 Line 1024: \RESULT{Figure: Pickup bar chart 3 groups fixed/adaptive-bef/aft}
 Line 1029: \RESULT{Table: Coordination summary 2-gen, 3PH fault ...}

A.3 Report Naming: Actual vs. Expected

Expected path	Expected content	Exists?	Status
sambp_sync_oc/report/	TR-01 Milestone 1	Yes	Complete PDF
sambp_sync_oc/report2/	TR-02 Milestones 2–5	Yes	Skeleton PDF
sambp_sync_oc/report3/	TR-03 Milestone 2 full	No	Missing
sambp_sync_oc/report4/	TR-04 Milestone 3	No	Missing
sambp_sync_oc/report5/	TR-05 Milestones 4+5	No	Missing
sambp/report3/	87T/87L Diff (pivot start)	Yes	Different track
sambp/report4/	87B Bus Diff	Yes	Different track
sambp/report5/	SAMBP integration	Yes	Different track

Note that sambp/report3/ through sambp/report55/ belong to the *main SAMBP project* (separate directory), not to the sambp_sync_oc/ sub-project. The numbering collision (sync_oc report3/ vs. main SAMBP report3/) never arose because sync_oc report3/ was never created.

B TR-02/2026 Four-Extension Summary

Extension	Core Mathematics	Script	Status
Ext 1 (M2) HIF	$\text{TMS}_{adapt} = \text{TMS}_{nom} \cdot f(I_{ss})$	run_milestone2.py	Formulated only
Ext 2 (Stage 2) Park dq	9-state ODE with AVR+PSS+Gov	main_run_case.py --stage 2	Formulated only
Ext 3 (M3) Sequence	Hilbert–Fortescue decomposition	main_run_case.py --case 1	Formulated only
Ext 4 (M4) Multi-gen	Greedy cascade coordination	run_multi_gen.py	Formulated only
